CS-412 Operating Systems
(Mid Semester Test I)

Date: March 13, 2026 **Time: 1.15 hr** **Marks: 20**

Answer all the questions. Each question carries 5 marks.

1. Why is the separation between user mode and kernel mode considered a fundamental principle of robust operating system design? Give an example that illustrate a user process being switched from user mode to kernel mode and then back to user mode.

2. Suppose we are the developers of an operating system that uses a five state process model i.e. new, ready, running, exit and blocked. In our OS, all processes execute in a round robin fashion. Further suppose that we are developing a new operating system for computers that are expected to have numerous I/O bound jobs. To ensure these jobs receive the processor quickly when needed (i.e. when a current I/O burst is finished), we are considering two READY states in our process model: the READY-CPU state (for CPU bound processes) and the READY-IO state (for I/O bound processes). Processes in the READY-IO state will be given the highest priority to access the CPU in the system and will execute in a FCFS fashion. Processes in the READY-CPU state will execute in a round robin fashion. Processes are classified as either CPU-bound or IO-bound when they enter the system.

   Discuss the pro/cons of implementing the OS with two ready states as mentioned here.

3. Consider the following five processes to run in a single CPU using preemptive priority scheduling. Show the scheduling order of the processes using Gantt chart. What is the average waiting time? (Smaller priority number indicates higher priority.)

Process	Arrival Time	Burst Time	Priority
P1	0	10	3
P2	1	4	1
P3	2	6	4
P4	3	5	2
P5	5	3	1

4. Consider a system of five philosophers sitting around a circular table. Each philosopher alternates between thinking and eating. A philosopher must acquire two forks (left and right) to eat. Each fork is represented by a binary semaphore. Write the semaphore based pseudocode for each philosopher assuming fork[i] is a binary semaphore initialized to 1. Analyse whether the given algorithm can lead to starvation or not.

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End Semester Examination, Winter Semester 2026
CS 412--Operating Systems

Time: 3 Hrs. **Date: May 18, 2026** **Marks: 60**

Note: Answer all the questions. Each question carries ($7^{1/2}+7^{1/2}=15$) marks.

1. (a) Explain how micro-kernel architecture supports extensibility and dynamic service replacement. Explain why efficient Inter Process Communication is considered the central challenge in micro-kernel design.

(b) What is the purpose of the system call **fork ()** in the UNIX operating system? Write a **C** routine to create a child process using the fork system call. Incorporate an error check in your routine in case the creation of the child process fails.

2. (a) In a multiprogramming operating system using dynamic partitioning, external fragmentation gradually increases over time. Critically evaluate a compaction strategy and discuss their impact on CPU utilization.
This diagram shows an example of memory configuration under dynamic partitioning, after a number of placement and swapping-out operations have been carried out. Addresses go from left to right; gray areas indicate blocks occupied by processes; white areas indicate free memory blocks. The last process placed is 2 Mbytes and is marked with an X. Only one process was swapped out after that.

	4M		1M	X	5M		8M		2M		4M		3M	

a. What was the maximum size of the swapped-out process?

b. What was the size of the free block just before it was partitioned by X?

c. A new 3-Mbyte allocation request must be satisfied next. Indicate the intervals of memory where a partition will be created for the new process under the following three placement algorithms: best-fit, first-fit and worst-fit. For each algorithm, draw a horizontal segment under the memory strip and label it clearly.

(b) Critically examine the necessity of paging the page tables in modern virtual memory systems. Why can page tables themselves become a major memory-management challenge in large address-space architectures? Explain how hierarchical paging addresses this issue, and analyze its trade-offs in terms of memory overhead, address translation latency, and system performance.

3. (a) Why does the process typically experience an extremely high page fault frequency at startup, and how does this phenomenon relate to the principles of locality of reference and working-set formation?

Consider the following page reference string:

6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 2, 0

Assuming demand paging with three frames, how many page faults would occur for the following page fault replacement algorithms?

- LRU replacement
- FIFO replacement
- Optimal page replacement

(b) Compare the performance of C-SCAN, SCAN and FCFS disk scheduling, assuming a uniform distribution of requests. How the directional movement of the disk head in SCAN and C-SCAN influences latency distribution compared to FCFS? Discuss why C-SCAN is often considered more suitable for time-sharing systems.

4. (a) How does the principle of least privilege enhances the protection of Operating System? How can systems that implement the principle of least privilege still have protection failures that leads to security violations. Explain with an example.

(b) Discuss the vulnerabilities inherent in fixed-priority multilevel queue systems and analyze how user behavior such as process classification manipulation, frequent I/O requests, or interactive-job simulation can unfairly influence CPU allocation.